Penetration Testing Report

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY & ETHICAL HACKING | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	**Chukwunonye Charles Ebuka**
Program / Batch	**B084-Networkwalks**
Date	**12 November 2026**
Modules Completed	**W2-PM1 (Multiple Kali tools)** **W2-PM2 (GHDB)** **W2-PM4 (theHarvester)** **W2-PM5 (Zenmap)**
Client / Target	**1. Networkwalks** **2. My LAN Network**
Permission secured from client?	**Yes**
Phases Covered	**Phase 1: Reconnaissance & Footprinting** **Phase 2: Scanning & Network Discovery**

1. LIABILITY DISCLAIMER

This penetration testing report was conducted strictly within authorized boundaries as part of the Networkwalks Cybersecurity Internship Programme (Batch B084). All reconnaissance and scanning activities were performed only against:
networkwalks.com (authorized target domain)
Local LAN subnet 192.168.56.0/24 (intern's own private lab environment)
No unauthorized systems were accessed. All findings are for educational purposes only. The intern accepts full responsibility for ensuring all activities remain within the defined scope.

2. INTRODUCTION

This report documents the footprinting, reconnaissance and network scanning activities performed during Week 2 of the Networkwalks Cybersecurity Internship. The objective was to simulate the information gathering phase of a penetration test using industry standard tools.
Modules completed:
W2-PM1: Footprinting & Reconnaissance with Multiple Kali Tools
W2-PM2: GHDB Google Hacking Database Footprinting
W2-PM4: theHarvester Footprinting
W2-PM5: Zenmap Network Scanning

3. TOOLS USED

Tools	Purpose
WHOIS	Domain registration lookup
WhatWeb	Web technology fingerprinting
NSLookup	DNS resolution
Curl -I	HTTP header inspection
Wafw00f	WAF detection
DNSRecon	DNS enumeration
Google Dorks/GHDB	Passive reconnaissance
theHarvester	Email and subdomain harvesting
Zenmap	Network scanning and host discovery

4. ACTIVITIES PERFORMED

W2-PM1 — Footprinting & Reconnaissance

I performed passive and semi-passive reconnaissance on networkwalks.com using six Kali Linux tools. Each tool targeted a different layer of the target's public infrastructure.

WHOIS — Queried domain registration records to identify ownership, registrar, creation/expiry dates and name servers.

WhatWeb — Fingerprinted web technologies, server type, CMS platform and plugins running on the target.

NSLookup — Resolved the domain to its public IP address via DNS.

Curl -I — Retrieved HTTP response headers to inspect server configuration, security policies and API paths.

Wafw00f — Detected whether a Web Application Firewall was protecting the target.

DNSRecon — Enumerated all DNS records including mail servers, name servers, SPF/TXT and SRV records.

W2-PM2 — GHDB Google Hacking Database

I accessed the Google Hacking Database on Exploit-DB and used specialized dork queries to identify publicly exposed information about the target. Dorks were executed using operators such as `site:`, `filetype:`, `inurl:` and `intitle:`. Results revealed indexed pages and potential login portals accessible via Google search.

W2-PM4 — theHarvester

I used theHarvester to harvest emails, subdomains and hosts from public sources. Two searches were run against microsoft.com — one using Baidu as the source and one using all available sources. The Baidu search returned no results. The all-sources search returned subdomains and hosts, with some sources requiring API keys.

W2-PM5 — Zenmap Network Scanning

I installed Zenmap on a Windows VM and ran a Ping Scan against my local subnet 192.168.56.0/24. The scan discovered 5 live hosts, identified their IP addresses, MAC addresses and vendors, and produced a network topology map which was saved as PDF.

5. RISK ANALYSIS

Finding	Tool	Impact	Risk Level
Domain registration details exposed	WHOIS	Attacker can identify registrar and contact info	Low
Web technologies fingerprinted	WhatWeb	Attacker knows exact software versions to target	Medium
DNS records fully enumerable	DNSRecon	Mail servers and infrastructure exposed	Medium
No WAF detected / WAF detected	Wafw00f	Affects difficulty of web attacks	Medium/Low
CCTV Login portals indexed by Google	Google Dorks (GHDB)	Direct attack surface for credential attacks	High
5 hosts discoverable via ping scan	Zenmap	Network topology visible to attacker on same network	Medium

6. RECOMMENDATIONS

Restrict WHOIS data

Use WHOIS privacy protection to hide registrant details.

Hide web technology headers

Configure server to remove version information from HTTP headers.

Implement WAF

If not already present, deploy a Web Application Firewall.

Remove sensitive pages from Google index

Use `robots.txt` and `noindex` tags to prevent login portals from being indexed.

Network segmentation

Ensure production hosts are not discoverable via simple ping scans.

DNS hardening

Restrict zone transfers and limit publicly exposed DNS records.

7. CONCLUSION

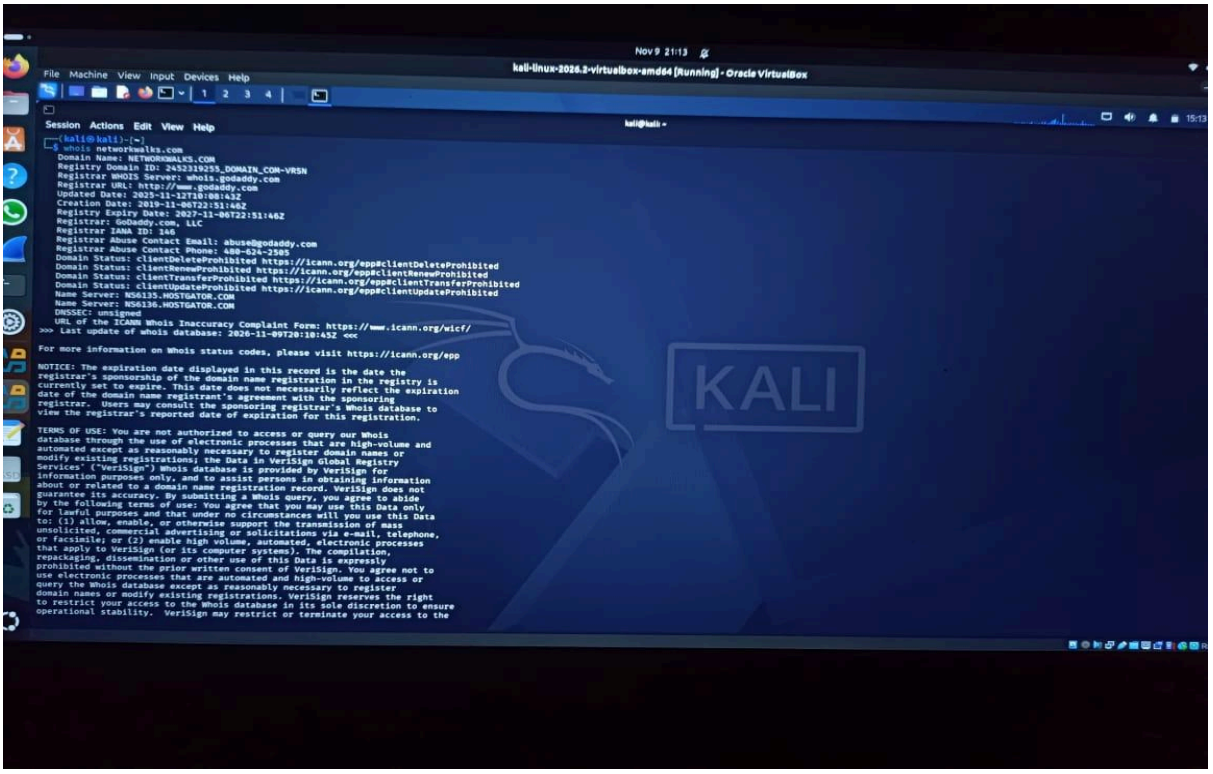
This week's modules demonstrated how much information is publicly available about any organization before a single packet is sent to their systems. Using only passive and semi-passive techniques, significant infrastructure details were gathered about networkwalks.com including DNS records, web technologies and HTTP headers.

The network scanning module demonstrated how quickly an attacker on the same network can map all live hosts, their IP addresses, MAC addresses and vendors. These findings highlight the importance of information security hygiene at every layer.

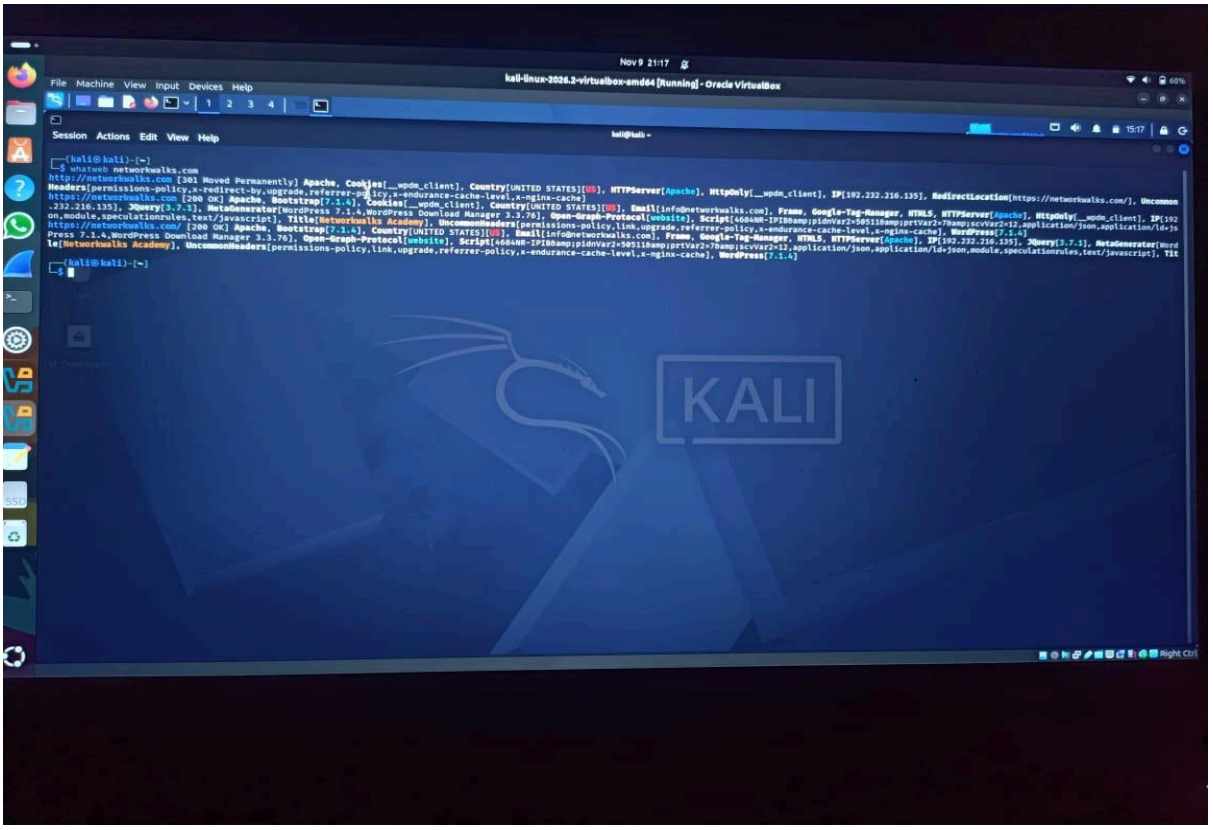
All activities were performed ethically within authorized scope as part of the Networkwalks Internship Programme.

8. Evidence

W2-PM1 — Footprinting & Reconnaissance



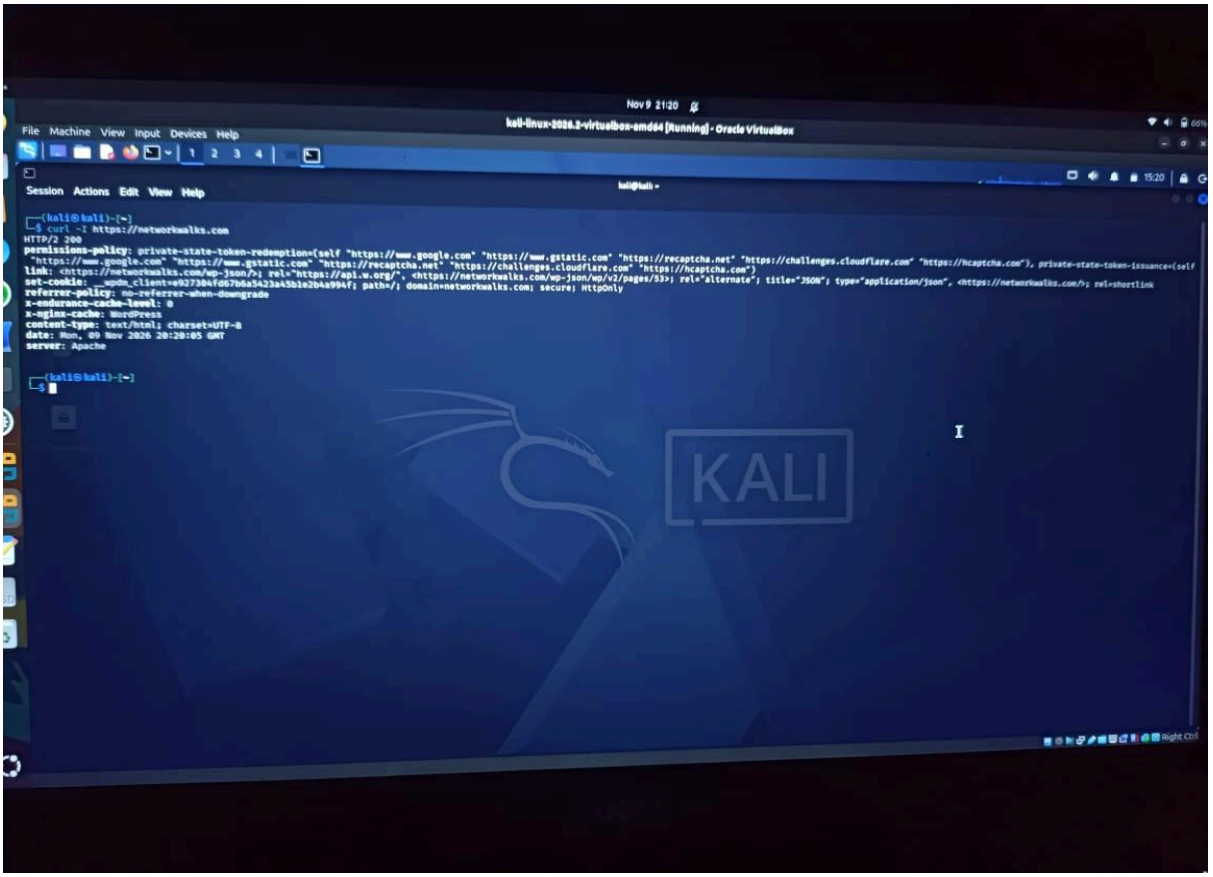
Screenshot 1 — WHOIS output showing domain registration details for networkwalks.com



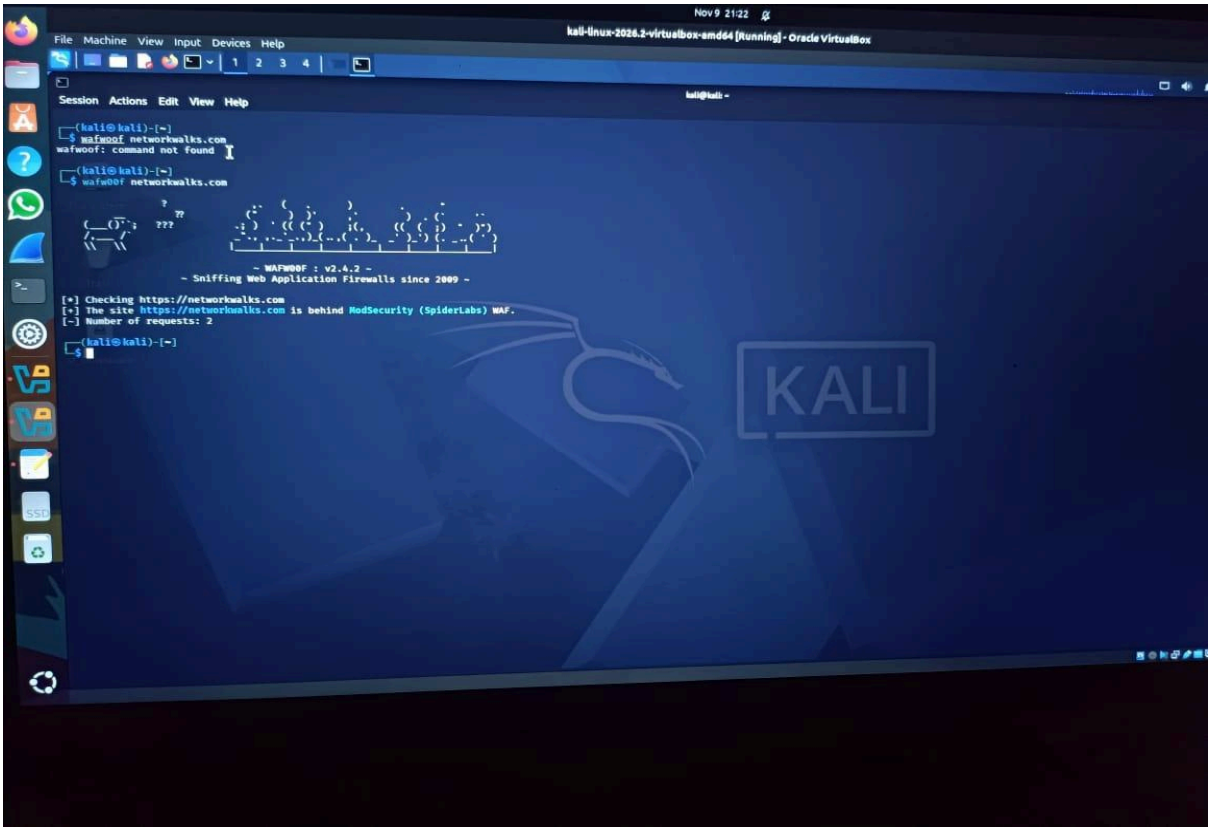
Screenshot 2 — WhatWeb output showing identified web technologies



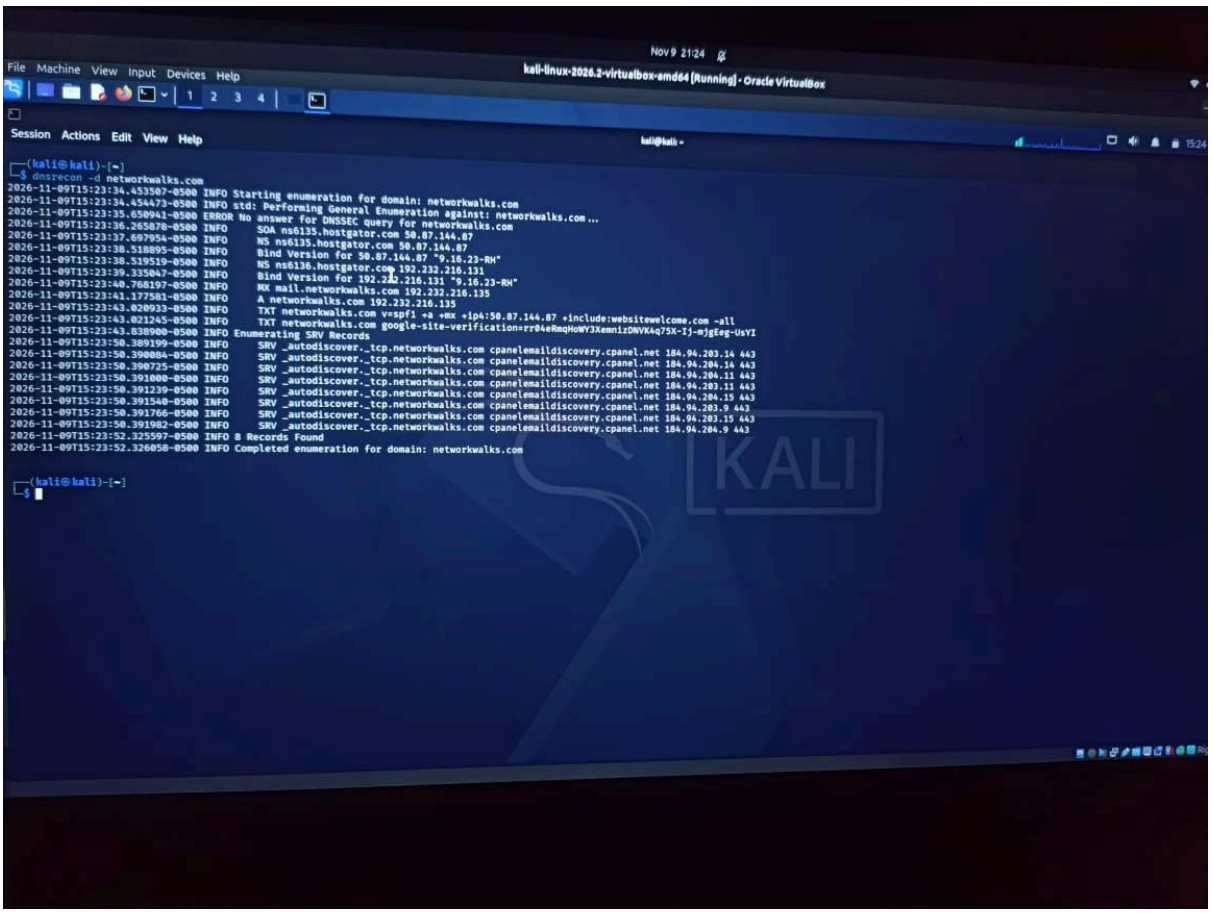
Screenshot 3 — NSLookup output showing resolved IP address



Screenshot 4 — Curl -I output showing HTTP response headers



Screenshot 5 — Wafw00f output showing WAF detection result



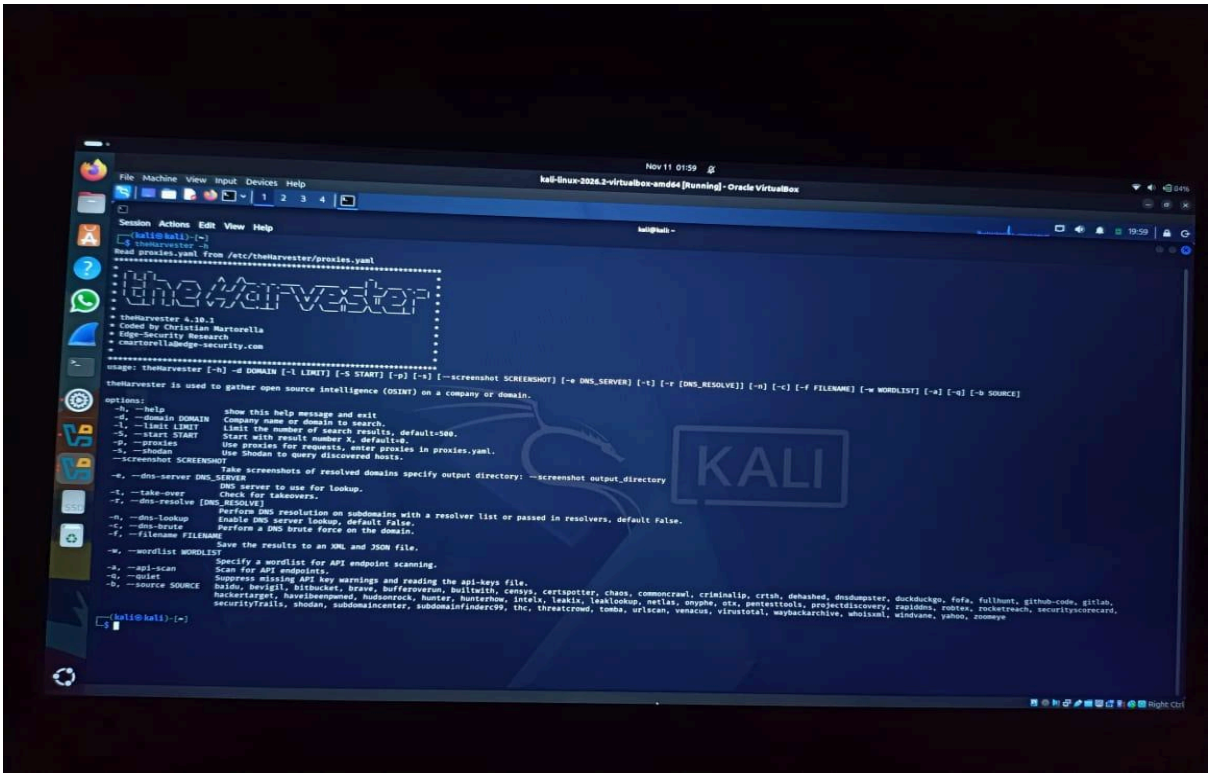
Screenshot 6 — DNSRecon output showing all DNS records.

W2-PM2 — GHDB Google Hacking Database



Screenshot 7 — Google dork results (sensitive IPs blurred)

W2-PM4 — theHarvester



Screenshot 8 — theHarvester help output

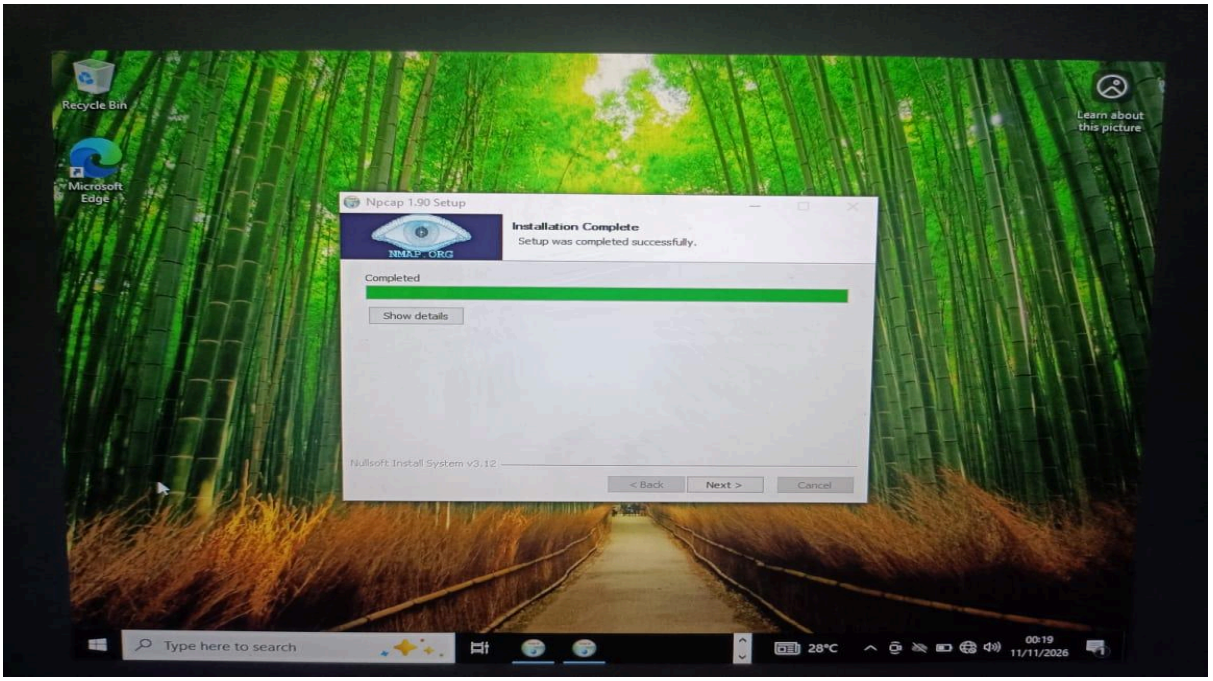


Screenshot 9 — Baidu search showing no results

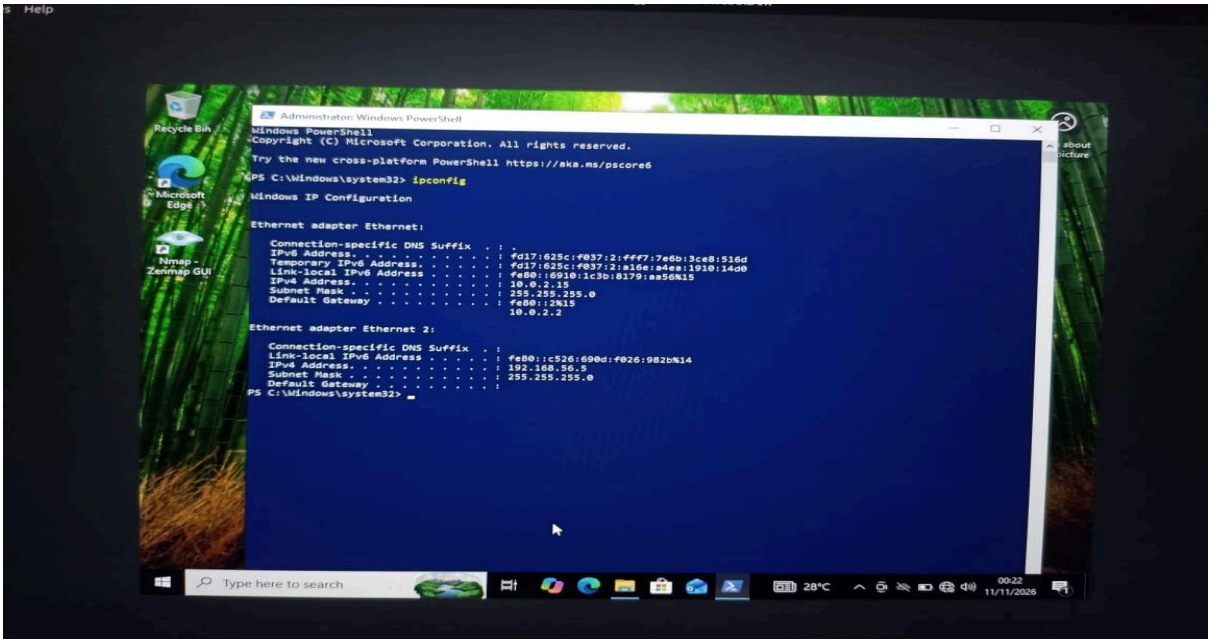


Screenshot 10 — All sources search showing results

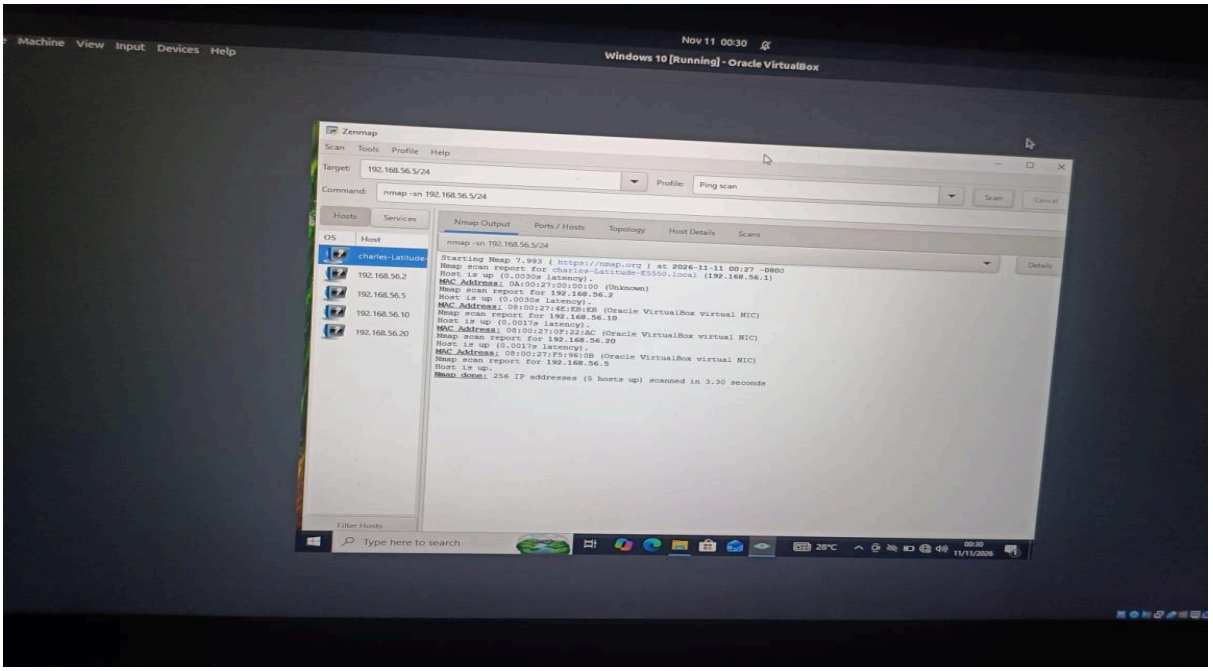
W2-PM5 — Zenmap



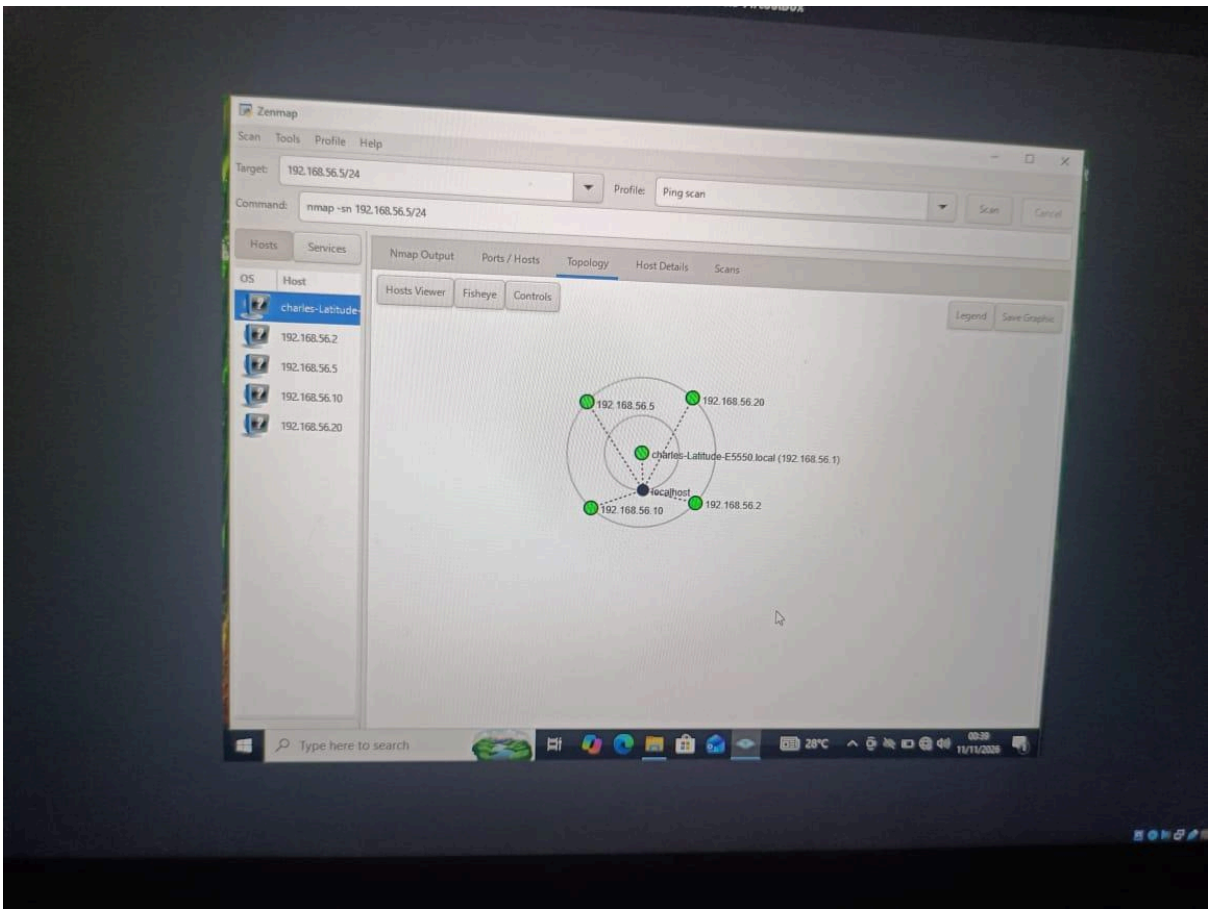
Screenshot 11 — Zenmap installation complete



Screenshot 12 — ipconfig showing local IP and subnet



Screenshot 13 — Ping scan results showing 5 live hosts



Screenshot 14— Network topology map